Software Requirements Specification (SRS)
Project Lane Management System

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1 Introduction

This software requirements specification document will cover all the requirements needed for the Lane Management System project for customer Ayush Agrawal in association with General Motors.

The contents that will be covered in this software requirements specification document are as follows:

1. Introduction
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    1.3. Definitions, acronyms, and abbreviations
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    2.3. User Characteristics
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    2.5. Assumptions and Dependencies
    2.6. Approportioning of Requirements
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    4.2. Domain Model
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1.1 Purpose

The purpose of the Software Requirements Specification document is to address validation criteria for software along with creating a clear and concise agreement between the customer and developer. This document serves as the foundation of why the software was developed and how it addresses client needs. Additionally, this document outlines how the software is intended to assist drivers who utilize the Lane Management System within their vehicle(s).

1.2 Scope

The Lane Management System is an embedded automotive software product that is composed of the Lane Departure Warning System, the Lane Keeping System, and the Lane Centering System.

The Lane Management System is a product to help assist drivers. The driver is still responsible for operating the vehicle and has the ability to override the corrections. The benefits to the Lane Management System are as follows:

- **Improving driver attentiveness:** The Lane Departure Warning System alerts drivers when they begin to drift outside of lane boundaries.
- **Preventing lane drifting caused collisions:** The Lane Keeping System will make steering adjustments when the vehicle is exiting the lane and the driver is unattentive, helping to bring the vehicle back into the correct lane.
- **Reducing driver fatigue:** The Lane Centering System takes over the repetitive task of maintaining lane position, making constant small adjustments to keep the vehicle centered.

The Lane Management System operates on vehicles that travel on roads that are properly marked as determined by the system. It determines that they are marked properly by using imagery from the vehicles cameras. The system then determines the vehicles position within the lane and warns the driver if there is a lane departure, and steers the vehicle back into the lane if driver does not respond and make a correction themself. The system also applies minor corrections to ensure the vehicle is near the center of the lane. The purpose of the Lane Management System is to correct unintentional lane departures and keep the car within a certain distance of the center of the lane.

The Lane Management System does not interact with speed, braking, or acceleration and doesn’t change lanes. The Lane Management System also does not detect people, obstacles or other vehicles. The Lane Management System is also only operable between 35 and 85 mph while also detecting properly marked lanes. If those conditions are not met then the system deactivates and notifies the driver.

1.3 Definitions, acronyms, and abbreviations

Camera Sensing Subsystem: The collection of cameras on the car that capture images and send them to the image processing subsystem.

Image Processing Subsystem: Processes received images and identifies lane markings.

Lane Centering System (LCS): Makes small adjustments to the steering wheel in order to keep the vehicle centered in the lane.

Lane Departure Warning System (LDWS): Displays visual and auditory warnings to the driver through the user interface when the vehicle leaves the lane.

Lane Keeping System (LKS): Makes larger adjustments to the steering wheel in order to stay within the lane. Considered an enhancement of LDWS.

Lane Management System (LMS): The software as a whole, with multiple systems and subsystems included within it.

Path Prediction Subsystem: Uses the information gathered by the Image Processing Subsystem and the Vehicle State Estimation System to predict the path of the vehicle and activate the LDWS, LCS, and LKS if necessary.

Supervisory Control System: Controls all of the subsystems, disables and enables them as necessary as well as provides diagnostics.

User Interface System: Displays warnings and notifications as necessary.

Vehicle State Estimation System: Calculates the speed of the vehicle, steering angle, and road curvature.

1.4 Organization

After section 1, the rest of the document is organized as follows:

Section 2 details high level topics relating to the LMS system. Product context, major functionality, user expectations, constraints, and assumptions made about operating conditions are discussed.

Section 3 includes an enumerated list of requirements that define the behavior, constraints, and invariants of the system.

Section 4 models the structure and behavior of the system in accordance with the requirements from section 3. The following models/diagrams are included:

- Use Case Diagram
- Domain Model
- Sequence Diagram
- State Diagram

Section 5 provides a prototype of the system to demonstrate the system behavior.

Section 6 provides resources used to research the domain and design the LMS.

Section 7 provides a point of contact to learn more about this document and project.

2 Overall Description

This section will cover the context for the product, the major functions that the software will perform, the expectations about the user, safety-critical properties and other constraints, and assumptions made about the hardware, software, environment, and user interactions.

2.1 Product Perspective

A key component of being a driver is being able to acknowledge differences in lanes and their purposes on the highway or public roads. However, not every environment that a driver will usually be in makes it easy to identify lane markings on the road. The implementation of a Lane Management System addresses safer driving by detecting lanes and properly adhering to road rules through lane detection. As the automotive industry continues to evolve, new features are being created to ensure safety and security for future drivers which allows for the expansion of Lane Management systems.

The Lane Management System is an umbrella system containing three subsystems of which are integrated to serve as lane management. The subsystems are Lane Keeping System, Lane Centering System and Lane Departure Warning System.

Interface Constraints:

- **System Constraints:** The Lane Management System will have to interact with other systems within the vehicle to be able to fully function. This includes being able to communicate with the vehicle's braking and steering system, along with bridging into other vehicle safety systems the vehicle may have in place outside of LMS.
- **User Interface:** The Lane Management System has to be able to directly notify the driver about changes or warnings within their driving. This may come in different forms but directly ties into the vehicle dashboard for visual warnings and notifications, the vehicle's steering wheel to feel the torque difference when any other subsystem under LMS is activated and the vehicle's audio system to be able to audibly notify the driver about changes or precautionary measures that should be taken.
- **Hardware Interface:** The Lane Management System must interact with the camera sensors within the vehicle to identify lane markings
- **Software Interface:** The Lane Management System works with software to maintain the camera sensor feedback, such as memory usage and data management of the given feedback
- **Communication Interface:** The Lane Management System ties into the visual

and audible notifications given off by the vehicle through the dashboard and audio system, providing communication for the driver

2.2 Product Functions

Lane Detection: The Lane Management System uses the vehicles cameras to capture images and processes them to determine where the vehicle is within the lane if it is deemed that the road markings can be detected properly.

Lane Departure Warning: If the vehicle is leaving the lane then the Lane Departure Warning System warns the driver on the dashboard and by vibrating the steering wheel. The warnings stop once the vehicle returns to its lane.

Lane Keeping: If the driver does not respond to the warning then the Lane Keeping System steers the driver back towards the center of its lane and prevents the driver from drifting into the other lane or off the road as long as the needed torque doesn't exceed the threshold.

Lane Centering: While the vehicle is in its lane, the LCS makes steering adjustments to keep it centered if it is outside the offset threshold.

Driver Override: If the driver uses a turn signal, then the LMS is deactivated and will not warn the driver or make any adjustments to the vehicle's position. Once the turn signal is disabled then the LMS is reactivated.

Logging: Every item the LMS records information or makes a decision the system will log it with a time stamp

2.3 User Characteristics

The user is expected to be qualified and able to operate the vehicle legally and safely. The user is also expected to have read the provided user manual and understand the functionality of the LMS.

2.4 Constraints

The LMS's behavior and operating conditions are constrained by the following:

1. The LMS requires an interface with the following subsystems to operate:
    a. Camera Sensing Subsystem
    b. Image Processing Subsystem
    c. Vehicle State Estimation System
    d. Path Prediction Subsystem
    e. User Interface System
    f. Supervisory Control Systems
2. Subsystems shall have the capability to operate in parallel with one another. For

example, the LDWS may be sending lane departure warnings while the LKS is sending steering input to correct the lane departure.

3. All events occurring within the LMS shall be logged, including: vehicle state, speed headings, driving conditions, what triggered the event, what the event was, all warnings, all overrides.
4. The LMS shall only operate at vehicle speed between 35 and 85 mph (exclusive).
5. The LMS shall only operate when a single lane can be detected from a pair of lane markers.

The LMS is a safety critical system. Meaning failure in certain functionalities of the system could incur serious injury or loss of life. To prevent this, additional security considerations have been made for the following properties:

- **Steering**: with the LMS, the vehicle receives steering input from the driver, the LKS, and the LCS. Multiple actors controlling the steering at once could produce dangerous behavior, so a hierarchy over control has been created. The driver has ultimate priority over steering, and any meaningful torque applied to the steering wheel from the driver will override steering input from either the LKS or LCS. Additionally, because the LKS is activated in response to lane departure, it has steering priority over the LCS.
- **Warnings**: when a lane departure has been detected, the driver must be warned sufficiently. To guarantee this, the LDWS warns the driver through three separate methods: visual, aural, and physical (vibration).

2.5 Assumptions and Dependencies

Some assumptions have been made about the hardware in use with the Lane Management System. The first is that the required hardware components will be present in every vehicle where the Lane Management System is implemented. These hardware components include camera sensors, steering wheel, and a turn signal. In addition it is assumed that the needed hardware to display warning signals to the driver is implemented in any vehicle where the Lane Management System is used such as ability to show warnings on the drivers dash board, ability to give haptic warning through the steering wheel and driver seat, and the ability to play auditorial warnings through the car speakers.

Some assumptions have been made about the software of the vehicle in use with the Lane Management System. The first is that the software for other systems such as steering and driver dashboard will have APIs that the Lane Management System will call to send signals to. Additionally, it is assumed that these software systems will work when these API calls are made, if the software systems are down the API calls will notify the Lane Management System as such.

One assumption has been made about the environment in which the Lane Management System is used. It is assumed that for the scope of this project the Lane Management System will operate under American road laws and regulations.

One assumption has been made about the user interactions with the Lane Management System. It is assumed the user is knowledgeable about the workings of the system, knowing how to override the system and turn it off.

2.6 Approportioning of Requirements

While the documentation has specific requirements that are addressed through a working Lane Management System, other requirements will arise that are to be addressed through future versions of the product. This includes understanding how the Lane Management System should react with other vehicles entering the current driver's lane, along with determining the speed threshold for camera visibility to work.

3 Specific Requirements

1. The Lane Management System shall consist of a Lane Departure Warning System, Lane Keeping System, and a Lane Centering System.
2. The Lane Management shall only be activated within a specific speed threshold.
    2.1. Attempts by the driver to activate the Lane Management System when the vehicle is outside of the speed threshold shall display an error message specifying the required speed threshold to the driver.
3. The Lane Management System shall be unable to make accurate decisions under the following circumstances, in these circumstances the Lane Management System shall be disabled.
    3.1. The road is too wide or too narrow.
    3.2. The lane markings are obscured in any way.
    3.3. The lane markings are faded.
    3.4. The lane markings are overly complicated.
    3.5. There are no lane markings.
4. If the Lane Management System is unable to make accurate decisions due to the failure of sensing devices or inaccurate sensing, the Lane Management System shall inform the driver with a visual and auditory alert.
    4.1. The visual warning shall be a notification that appears on the drivers dashboard.
    4.2. The audio warning shall be a beep played by the car's speakers.
    4.3. If the driver attempts to activate the Lane Management System when any of these faults are present the Lane Management System shall display a warning with the relevant diagnostic message on the driver's dashboard.
5. When the Lane Management System is shut down purposely the Lane Management System shall alert the driver with a visual alert.
    5.1. The visual warning shall be a notification that appears on the drivers dashboard.
6. The Lane Management System shall capture images from the cameras every predetermined amount of milliseconds/seconds to capture and/or calculate:
    6.1. The lane markings.

6.2. The road curvature.
6.3. The position of the car in the future.
6.4. The car’s location in the lane.

7. The Lane Management System shall keep a log of all decision making data.
    7.1. Logs shall contain the car's position in the lane, car speed, Lane Management System active status, alerts sent and the times they were sent, and commands sent to the steering system and the times they were sent.
8. The Lane Management System shall be able to be overridden by the driver’s steering at any point in operation.
9. The Lane Management System shall be deactivated when either turn signal is switched on.
10. The Lane Management System shall be deactivated when the driver applies significant torque on the steering wheel.
11. The Lane Management System shall be reactivated when the turn signal is switched off if it was active before the turn signal was switched on.
12. The Lane Management System shall have a way of grading the image capture quality.
    12.1. The Lane Management System shall have a threshold of image quality to pass in order to stay active.
13. The driver shall be able to disable and enable the Lane Management System.
    13.1. The System shall notify and alert the driver with an auditory alert each time the system is enabled
    13.2. The Lane Management System’s current state shall be displayed on the dashboard
14. Lane Departure Warning System
    14.1. When the driver exits their lane they shall receive a warning consisting of a visual, audio, and physical component.
        14.1.1. The visual warning shall be a notification that appears on the drivers dashboard.
        14.1.2. The audio warning shall be a beep played by the car's speakers.
        14.1.3. The physical warning shall be vibration of the seat and the steering wheel.
    14.2. The Lane Departure Warning System shall be able to notify the driver that it is out of the lane after x seconds.
    14.3. The Lane Departure Warning System shall turn off the warning once the driver is back in their lane.
    14.4. Repeated infractions shall result in warning the driver of the danger of driving distracted and/or driving tired
15. Lane Keeping System
    15.1. If no driver action is taken after the Lane Departure Warning System activates the Lane Keeping System shall gently steer the car back to the center of the lane.
    15.2. Lane Keeping System shall prevent drivers from drifting off the road.
    15.3. When the Lane Departure Warning System is inoperable the Lane Keeping System shall be inoperable.
16. Lane Centering System

16.1. The Lane Centering System shall continuously check lane markings to detect where the car is relative to the lane.
16.2. The Lane Centering System shall send steering commands to keep the vehicle's path centered within the detected lane.

17. Cybersecurity
    17.1. Threat vectors: the messages that are exchanged between the LMS and the vehicle subsystem as well as the camera imagery and logs stored from all actions
    17.2. Threat actors: An attacker with access to the vehicles network, a vehicle owner that is attempting to bypass limits in requirements 2 or 3, and an attacker that is manipulating roadway markings or camera input
    17.3. The LMS should authenticate every message received from each other vehicle subsystem and reject any message that fails that authentication and also should reject ones that timestamps don't fit within a certain window of recency
    17.4. The LMS shall cross check vehicle position taken from imagery against the vehicle state estimation system data and treat a disagreement past a certain threshold due to inaccurate sensing
    17.5. The LMS shall validate every steering command against its range before sending it to the steering system
    17.6. The Lane Management system shall log every rejected message

4 Modeling Requirements

4.1 Use Case Diagram

Figure 1 shows the use case diagram created for the LMS. The use case diagram was created to show how the separate use cases are divided between the LDWS, LKS, and LCS as well as the actors responsible for each case. In use case diagrams, the actors (stick figures) connect to ovals which represent the use cases. Multiple actors can point to a single use case. These use cases can include, or extend other use cases. Include's are prerequisites that are often repeated between multiple use cases. Extends are exceptions for use cases such as error messages or overrides.

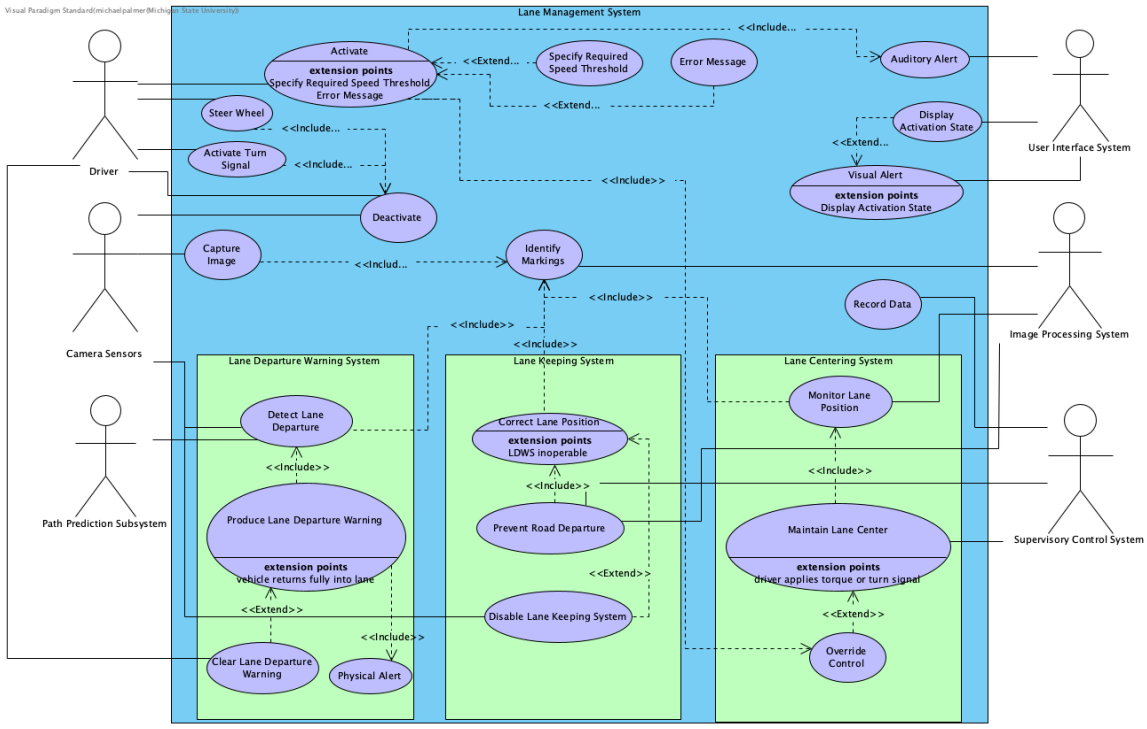


Figure 1: Use case model for the Lane Management System.

Use Cases Descriptions:

Use Case:	Activate
Actors:	Driver
Description:	The driver activates the Lane Management System. Perform Error Message if the car's current speed is outside of the required speed threshold. Perform Specify Required Speed Threshold if the car's current speed is outside of the required speed threshold.
Type:	Primary and Essential
Includes:	Auditory Alert
Extends:	N/A
Cross-refs:	2, 2.1, 13
Use cases:	N/A

Use Case:	Deactivate
Actors:	Camera Sensors, Driver
Description:	The camera sensors deactivate the Lane Management System if ● The road is too wide or too narrow. ● The lane markings are obscured in any way. ● The lane markings are faded.

	● The lane markings are overly complicated. ● There are no lane markings. Additionally, the driver can disable the lane management system. An auditory alert plays following this action.
Type:	Primary and Essential
Includes:	Auditory Alert
Extends:	N/A
Cross-refs:	3, 3.1, 3.2, 3.3, 13, 13.1
Use cases:	N/A

Use Case:	Specify Required Speed Threshold
Actors:	User Interface Subsystem
Description:	Shows the required speed threshold to the driver on the dashboard.
Type:	Secondary and Essential
Includes:	N/A
Extends:	Activate
Cross-refs:	2.1
Use cases:	The driver must have initiated the Activate use case.

Use Case:	Error Message
Actors:	User Interface Subsystem
Description:	Shows an error message to the driver on the dashboard.
Type:	Secondary and Essential
Includes:	N/A
Extends:	Activate
Cross-refs:	2.1
Use cases:	The driver must have initiated the Activate use case.

Use Case:	Capture Image
Actors:	Camera Sensing Subsystem
Description:	Captures images to be used by the image processing subsystem in identifying lane markers
Type:	Primary and essential
Includes:	N/A
Extends:	N/A
Cross-refs:	6
Use cases:	Identify Lane Markers

Use Case:	Identify Lane Markers
Actors:	Image Processing Subsystem
Description:	Identifies lane markers from given images produced by the Camera Sensing Subsystem
Type:	Primary and essential
Includes:	N/A
Extends:	N/A
Cross-refs:	6
Use cases:	Correct Lane Position

Use Case:	Record Data
Actors:	Supervisory Control System
Description:	Record decision making data to a collective log including the car's position in the lane, car speed, Lane Management System active status, alerts sent and the times they were sent, and commands sent to the steering system and the times they were sent.
Type:	Secondary
Includes:	N/A
Extends:	N/A
Cross-refs:	7
Use cases:	N/A

Use Case:	Override Control
Actors:	Driver, Supervisory Control Systems
Description:	The driver overrides the Lane Management Systems movement of the steering wheel
Type:	Primary
Includes:	N/A
Extends:	Correct lane position
Cross-refs:	8
Use cases:	N/A

Use Case:	Steer Wheel
Actors:	Driver
Description:	The driver adjusts the vehicle’s path by turning the steering wheel
Type:	Primary
Includes:	Deactivate
Extends:	N/A
Cross-refs:	10

Use cases:	N/A

Use Case:	Activate Turn Signal
Actors:	Driver
Description:	The driver activates either the left or right turn signal to indicate intent to merge into corresponding lane
Type:	Secondary
Includes:	Deactivate
Extends:	N/A
Cross-refs:	9, 11
Use cases:	N/A

Use Case:	Display Activation State
Actors:	User Interface System
Description:	While the lane management system is active, a symbol is displayed on the screen.
Type:	Secondary
Includes:	N/A
Extends:	N/A
Cross-refs:	13.1
Use cases:	N/A

Use Case:	Produce Lane Departure Warning
Actors:	Path Prediction Subsystem
Description:	A lane departure is detected. An auditory alert, visual, and physical alert is sent to warn the driver.
Type:	Primary
Includes:	Auditory Alert, Visual Alert, Physical Alert
Extends:	N/A
Cross-refs:	14.1
Use cases:	N/A

Use Case:	Auditory Alert
Actors:	User Interface System
Description:	A beep is played over the vehicle’s speaker systems
Type:	secondary
Includes:	N/A
Extends:	N/A
Cross-refs:	13.1, 14.1, 14.1.2

Use cases:	Produce Lane Departure Warning

Use Case:	Visual Alert
Actors:	User Interface System
Description:	Display message with relevant information on the screen
Type:	secondary
Includes:	N/A
Extends:	N/A
Cross-refs:	14.1, 14.1.1
Use cases:	Produce Lane Departure Warning

Use Case:	Physical Alert
Actors:	User Interface System
Description:	Produce a physical alert in the form of seat and steering wheel vibration
Type:	secondary
Includes:	N/A
Extends:	N/A
Cross-refs:	14.1, 14.1.3
Use cases:	Produce Lane Departure Warning

Use Case:	Lane Departure Detection
Actors:	Camera Sensors
Description:	Determines that the vehicle has crossed lane boundary and has remained outside of the lane for x seconds, which it then triggers the Lane Departure Warning System to alert the driver
Type:	Primary and Essential
Includes:	Identify Markings
Extends:	N/A
Cross-refs:	14.2
Use cases:	Detect Lane Departure, Physical Alert, Clear Lane Departure Warning

Use Case:	Clear Lane Departure Warning
Actors:	Camera Sensors, Image Processing System, Driver
Description:	Turns off the visual, audio, and physical lane departure warnings once the vehicle is fully inside its lane. Additionally, when the turn indicator is on, the lane departure warning system should not alert the driver
Type:	Primary and Essential

Includes:	Identify Markings
Extends:	Detect Lane Departure
Cross-refs:	9, 14.3
Use cases:	N/A

Use Case:	Correct Lane Position
Actors:	Supervisory Control System, Driver
Description:	When the driver takes no action after the Lane Departure Warning System Activates, the Lane Keeping System gently steers the vehicle back to the center of the lane
Type:	Primary and Essential
Includes:	Identify Markings
Extends:	Detect Lane Departure
Cross-refs:	15.1
Use cases:	Identify Markings, Disable Lane Keeping System

Use Case:	Prevent Road Departure
Actors:	Supervisory Control System, Image Processing System
Description:	Applies steering correct to keep the vehicle from drifting out of their lane when the outside of their boundary is the edge of the road
Type:	Primary and Essential
Includes:	Correct Lane Position
Extends:	N/A
Cross-refs:	15.2
Use cases:	Correct Lane Position

Use Case:	Disable Lane Keeping System
Actors:	Camera Sensors
Description:	Renders the Lane Keeping System inoperable whenever the Lane Departure Warning System is inoperable
Type:	Secondary
Includes:	N/A
Extends:	Correct Lane Position
Cross-refs:	15.3
Use cases:	N/A

Use Case:	Monitor Lane Position
Actors:	Camera Sensors, Image Processing System

Description:	Continuously evaluates the identified lane markings to determine the vehicles lateral position relative to detected line
Type:	Primary and Essential
Includes:	Identify Markings
Extends:	N/A
Cross-refs:	16.1
Use cases:	Identify Markings

Use Case:	Maintain Lane Center
Actors:	Supervisory Control System
Description:	Sends continuous steering commands to steering system to keep vehicles path centered within the detected lane
Type:	Primary and Essential
Includes:	Monitor Lane Position
Extends:	N/A
Cross-refs:	16.2
Use cases:	Monitor Lane Position, Override Control

Use Case:	Overly Complicated Lane Markings
Actors:	Car sensors
Description:	The disabling of LMS based on lane marking visibility
Type:	Primary (essential)
Includes:	N/A
Extends:	The Lane Management System shall be unable to make accurate decisions under the following circumstances, in these circumstances the Lane Management System shall be disabled.
Cross-refs:	3.4
Use cases:	N/A

Use Case:	No Lane Markings
Actors:	Car Sensors
Description:	The disabling of LMS based on no lane markings being visible
Type:	Primary (essential)
Includes:	N/A
Extends:	The Lane Management System shall be unable to make accurate decisions under the following circumstances, in these circumstances the Lane Management System shall be disabled.
Cross-refs:	3.5
Use cases:	N/A

Use Case:	Visual Warning Notification on Dashboard

Actors:	Car Sensors
Description:	If LMS becomes unable to make decision, notify driver
Type:	Primary (essential)
Includes:	4.2
Extends:	If the Lane Management System is unable to make accurate decisions due to the failure of sensing devices or inaccurate sensing, the Lane Management System shall inform the driver with a visual and auditory alert.
Cross-refs:	4.1
Use cases:	4.2,4.3

Use Case:	Audible Warning Notification on Dashboard
Actors:	Car Sensors
Description:	If LMS becomes unable to make decision, notify driver through audible warning
Type:	Primary (essential)
Includes:	4.3
Extends:	If the Lane Management System is unable to make accurate decisions due to the failure of sensing devices or inaccurate sensing, the Lane Management System shall inform the driver with a visual and auditory alert.
Cross-refs:	4.2
Use cases:	4.1,4.3

Use Case:	Visual Diagnostic Notification Warning Notification on Dashboard
Actors:	Car Sensors, Driver
Description:	If driver attempts to activate LMS, diagnostic message shall appear
Type:	Primary (essential)
Includes:	N/A
Extends:	If the Lane Management System is unable to make accurate decisions due to the failure of sensing devices or inaccurate sensing, the Lane Management System shall inform the driver with a visual and auditory alert.
Cross-refs:	4.3
Use cases:	4.1,4.2

Use Case:	Visual Warning Notification on Dashboard when LMS is purposely shut down
Actors:	Car Sensors, Driver
Description:	If LMS is purposely shut down, LMS shall alert driver with visual notification

Type:	Primary (essential)
Includes:	N/A
Extends:	When the Lane Management System is shut down purposely the Lane Management System shall alert the driver with a visual alert.
Cross-refs:	5.1
Use cases:	N/A

Use Case:	Capture Image
Actors:	Camera Sensing Subsystem
Description:	Captures images to be used by the image processing subsystem in identifying lane markers
Type:	Primary and essential
Includes:	None
Extends:	None
Cross-refs:	Requirement #6
Use cases:	Identify Lane Markers

Use Case:	Identify Lane Markers
Actors:	Image Processing Subsystem
Description:	Identifies lane markers from given images
Type:	Primary and essential
Includes:	None.
Extends:	None.
Cross-refs:	Requirement #6
Use cases:	Correct Lane Position

Use Case:	Record Data
Actors:	Supervisory Control System
Description:	Record decision making data to a collective log including the car's position in the lane, car speed, Lane Management System active status, alerts sent and the times they were sent, and commands sent to the steering system and the times they were sent.
Type:	Secondary
Includes:	None
Extends:	None
Cross-refs:	Requirement #7
Use cases:	None.

Use Case:	Override Control
Actors:	Driver, Supervisory Control Systems
Description:	The driver overrides the Lane Management Systems movement of the steering wheel
Type:	Primary
Includes:	None.
Extends:	Correct lane position
Cross-refs:	Requirement #8
Use cases:	None.

Use Case:	Disable Warning Signal
Actors:	None
Description:	When the turn indicator is on, the lane departure warning system should not alert the driver
Type:	Secondary
Includes:	None
Extends:	Alert driver
Cross-refs:	Requirement #9
Use cases:	None

4.2 Domain Model

A domain model was created to define the main elements of the LMS, attributes and major operations of each element, and the structure of connections between the elements. In the model, system elements are represented by boxes. Attributes and operations pertaining to an element are listed in its corresponding box. Lines between boxes denote some connection between elements, and the text attached to the line describes the nature and multiplicity of the connection. Lines with a black diamond on one end are a type of connection that represent that the element on the flat side of the line is a “part of” the element on the black diamond side. The completed domain model is shown below in Figure 2.

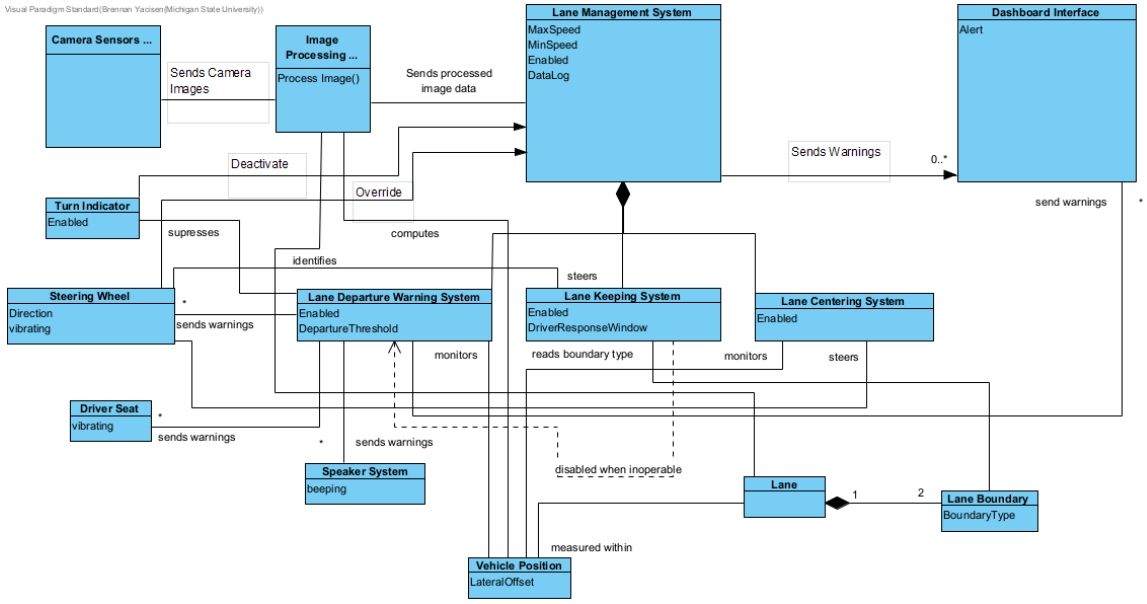


Figure 2: A domain model of the LMS system.

A data dictionary was created to organize the data shown in the domain model. The data dictionary is defined in Table 1 below.

Table 1: Data dictionary defined by domain model from Figure 2.

Element Name		**Description**
Camera Sensors Interface		Interface for the Lane Management System to communicate with the camera sensors.
Attributes		
Operations		
Relationships		Sends camera images to the Lane Management System.
UML Extensions		

Element Name		**Description**
Dashboard		Interface for the Lane Management System to communicate with the drivers dashboard.
Attributes	Alert: String	Message displayed to interface
Operations		
Relationships		Receives warnings to display from the Lane Management System.

	Receives lane departure warnings from the Lane Departure Warning System.
UML Extensions	

Element Name		**Description**
Driver Seat		The seat in which the vehicle driver sits
Attributes		
	Vibrating: Boolean	Enables vibration to be produced
Operations		
Relationships		Vibration setting enabled by the Lane Departure Warning System
UML Extensions		

Element Name		**Description**
Image Processing Subsystem		Processes the images gathered by the cameras and identifies lane markers
Attributes		
Operations		
	Process image	The operation to gather relevant data to be used by other systems
Relationships		Sends processed data to the Lane Management System. Receives camera images from the Camera Sensors. Receives Lateral Offset from Vehicle Position. Receives lane data from Lane.
UML Extensions		

Element Name		**Description**
Lane Centering System		A component of the Lane Management System. Continuously monitors the vehicle's position within the lane and steers the vehicle to maintain a centered position.
Attributes		
	Enabled: Boolean	True if the Lane Centering System is enabled
Operations		
Relationships		A component of the Lane Management System. Send steering signals to the steering wheel. Monitors Lateral Offset from Vehicle Position.
UML Extensions		

Element Name		**Description**
Lane Departure Warning System		A component of the Lane Management System. Warns the driver if they are drifting from the lane they are currently in.
Attributes		
	Enabled: Boolean	If the Lane Departure Warning System is currently enabled
Operations		
Relationships		A component of the Lane Management System. Sends warning signals to the Steering Wheel. Is able to be suppressed by the Turn Indicator. Disables the Lane Keeping System when inoperable. Sends warnings to the Dashboard. Monitors Vehicle Position. Sends warnings to the Driver Seat.

	Sends warnings to the Speaker System.
UML Extensions	

Element Name		**Description**
Lane Keeping System		A component of the Lane Management System. Steers the vehicle back into the lane if warning from the Departure Warning System is ignored.
Attributes		
	Enabled: Boolean	True if Lane Keeping system is enabled or disabled
Operations		
Relationships		A component of the Lane Management System. Sends steering signals to the Steering Wheel. Reads boundary type of Lane Boundaries. Is disabled when the Lane Departure Warning System is disabled.
UML Extensions		

Element Name		**Description**
Lane Management System		The Lane Management System is composed of the Lane Departure Warning System, Lane Keeping System, and Lane Centering System. The Lane management system coordinates interactions between all three subsystems and outside systems.
Attributes		
	MaxSpeed: Integer	The max vehicle speed in which the Lane Management System will be activated.
	MinSpeed: Integer	The minimum vehicle speed in which the Lane Management System will be activated.
	Enabled: Boolean	True if the Lane Management System is currently enabled.
	DataLog:	Records a log of all decision making data

	List	
Operations		
Relationships		The composite of the Lane Departure Warning System, Lane Keeping System, and Lane Centering System. Receives camera images from the Camera Sensors Interface. Sends warnings to be displayed to the Dashboard Interface. Can be deactivated by the Turn Indicator. Can be overridden by the Steering Wheel.
UML Extensions		

Element Name		**Description**
Speaker System		Interfaces with the speaker system in the vehicle.
Attributes		
	Beeping: Boolean	Enables production of warning sound
Operations		
Relationships		Beeping setting enabled by Lane Departure Warning System
UML Extensions		

Element Name		**Description**
Steering Wheel		The steering wheel controlled by the driver
Attributes		
	Direction: String	Identifies in which direction the steering wheel is being steered
	Vibrating: boolean	Enables vibration to be produced
Operations		

Relationships	Overrides the Lane Management System. Vibration setting enabled by Lane Departure Warning System. Can receive steering signals from the Lane Keeping System. Can receive steering signals from the Lane Centering System.
UML Extensions	

Element Name		**Description**
Turn indicator		The left and right turn indicator
Attributes		
	Enabled: Boolean	True if the turn indicator is enabled.
Operations		
Relationships		Disables the Lane Management System while the turn indicator is active. Disables the Lane Departure Warning System while the turn indicator is active.
UML Extensions		

Element Name		**Description**
Lane		The lane the vehicle is currently traveling in identified by the Image Processing Subsystem
Attributes		
Operations		
Relationships		Composed of two lane boundary objects. Identified by the Image Processing Subsystem. Used by Vehicle Position to produce lane positioning measurements.

UML Extensions	

Element Name		**Description**
Lane Boundary		One side of the detected lane The boundary type is what determines a painted marking from edge of road
Attributes		
	boundaryType: String	Solid, dashed, or road edge
Operations		
Relationships		Part of a Lane Read by Lane Keeping System
UML Extensions		

Element Name		**Description**
Vehicle Position		Position the vehicle is in within detected lane
Attributes		
	lateralOffset: Real	Distance from center of the lane, zero means centered
Operations		
Relationships		Computed by image processing subsystem. Measured relative to lane. Monitored by Lane Departure Warning System and Lane Centering System.
UML Extensions		

4.3 Sequence Diagrams

Sequence diagrams were made to aid in the visualization of normal and exceptional cases. The following diagrams cover a wide range of system behavior.

Figure 3 shows how the steering wheel deactivates the Lane Management System. First the driver turns the steering wheel. If the torque of the steering wheel is above the threshold to be considered purposeful the steering wheel sends a signal to the Lane Management System to deactivate. The Lane Management System distributes this signal to the three Lane Management System subsystems, the Lane Departure Warning System, the Lane Keeping System, and the Lane Centering System. All three of these subsystems deactivate in response.

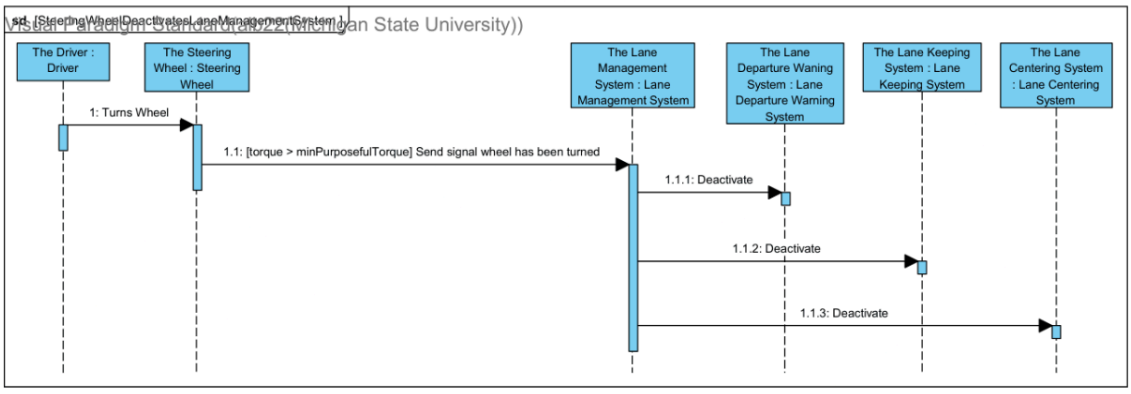


Figure 3: A sequence diagram showing the steering wheel deactivating the Lane Management System.

Figure 4 shows the exceptional case when the steering correction limit has been reached. The sequence starts with the image processing subsystem requesting an image from the camera sensing subsystem. The camera sensing subsystem sends an image in return. This image is sent to the path predication subsystem which determines the road curvature from the image and passes this data to the Lane Centering System. The Lane Centering System will then compute the required steering angle in order for the vehicle to maintain center lane position. The Lane Centering System will attempt to apply steering correction up to the maximum steering angle allowed. If the steering angle limit is reached and the vehicle is still outside of the lane center the Lane Centering System will send a warning to the driver to take control of the vehicle and then deactivate.

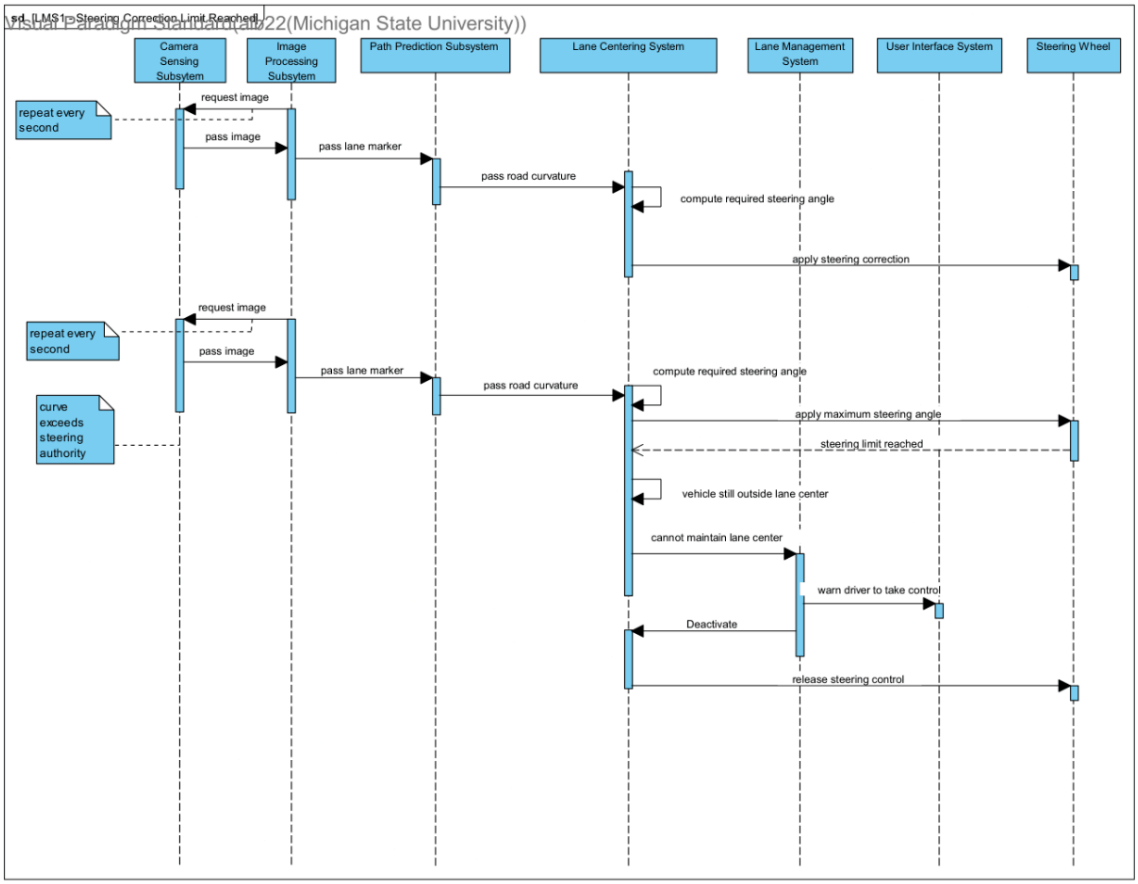


Figure 4: A sequence diagram showing the steering correction limit reached.

Figure 5 shows how the Lane Management System handles seeing multiple lane markings on the road. The sequence starts by the image processing subsystem requesting an image from the camera sensing subsystem. The camera sensing subsystem returns an image in response. The image processing subsystem determines the lane boundaries from this image and sends them to the Lane Management System. The Lane Management system cannot handle more lane boundaries than expected, due to this expectation the system will deactivate. To deactivate the Lane Management System sends deactivate signals to each of its three subsystems, the Lane Departure Warning System, the Lane Keeping System, and the Lane Centering System. All three of these subsystems deactivate in response.

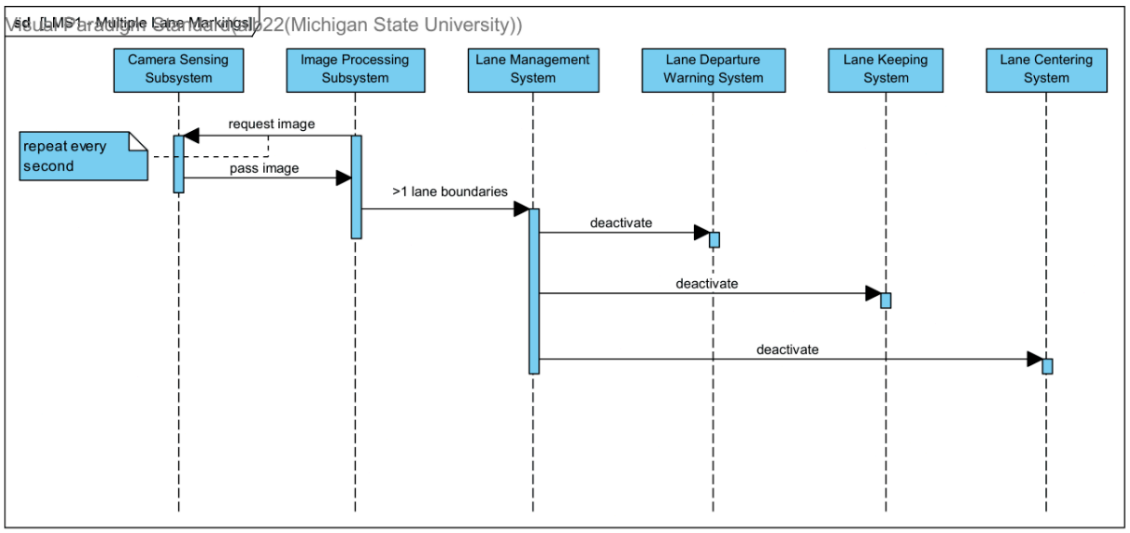


Figure 5: A sequence diagram showing the exception handling of the camera sensors detecting multiple lane markings.

Figure 6 shows how the Lane Centering System handles centering the vehicle in a normal case. The sequence begins with the image processing subsystem requesting an image from the camera sensing subsystem. The camera sensing subsystem sends an image to the image processing subsystem in return. The image processing subsystem processes the image to get the lane marker data and passes this information to the path prediction subsystem. The path prediction subsystem uses the lane marker data to predict the vehicles next position, it then sends this data to the Lane Centering System. The Lane Centering System computes the lane center offset and applies the needed steering correction to keep the vehicle in the center of the lane to the steering wheel.

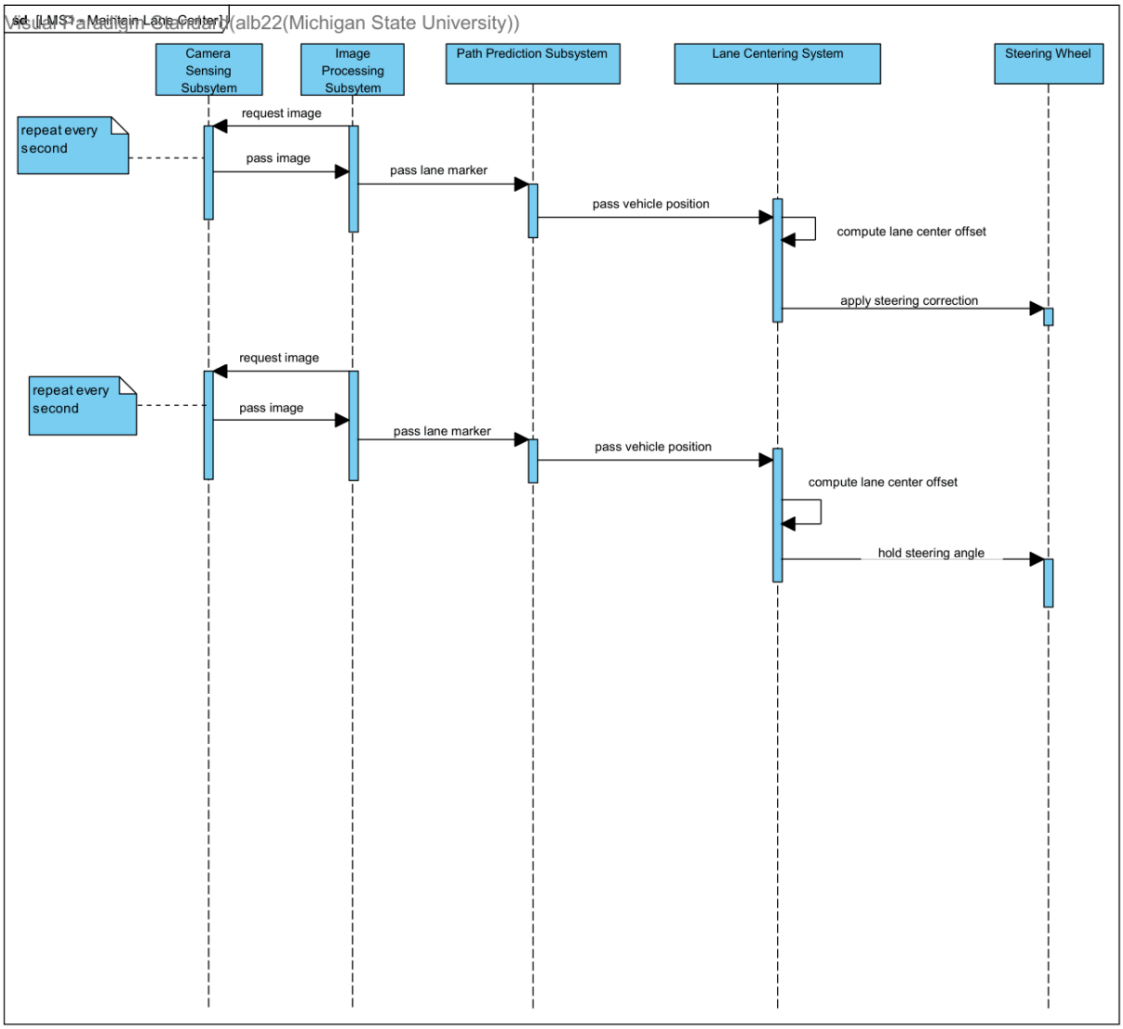


Figure 6: A sequence diagram showing the Lane Centering system maintaining the vehicles center lane position.

Figure 7 shows the Lane Departure Warning System and the Lane Keeping System handling a lane departure. First the image processing system requests an image from the camera sensing subsystem. The camera sensing subsystem sends an image back to the image processing system in return. The image processing system processes the images and sends lane marker data from them to the path prediction subsystem. The path prediction system will either detect a lane departure or not. If a lane departure is detected the path prediction system will signal a departure is detected to the Lane Departure Warning System and the Lane Keeping System. The Lane Departure Warning System will then warn the driver of the departure using the user interface system. The Lane Keeping System will apply torque to the steering wheel as needed to bring the vehicle back into the lane. Once a lane departure is not detected a signal will be sent to the Lane Keeping System to stop torque on the steering wheel.

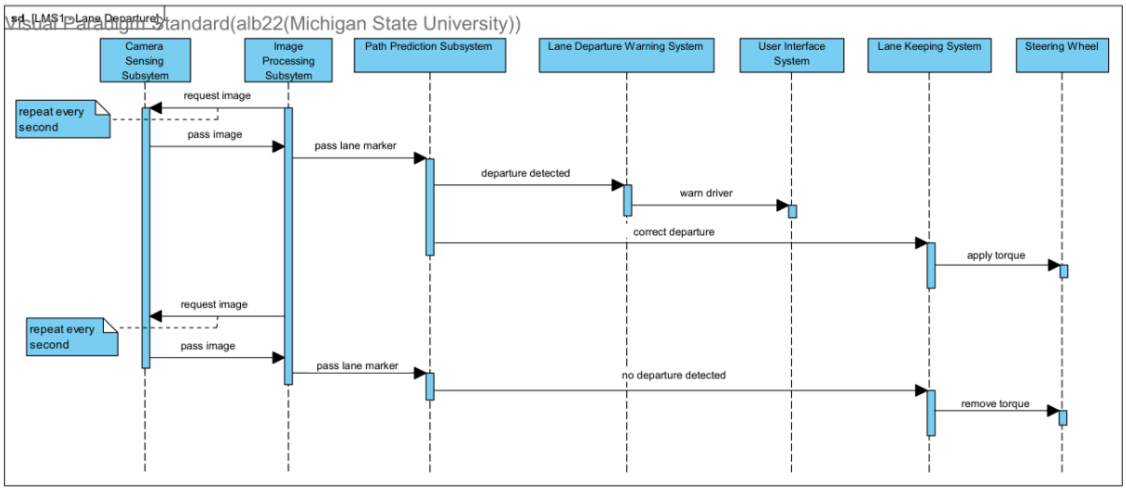


Figure 7: A sequence diagram showing the Lane Departure Warning System and the Lane Keeping Systems handling of a lane departure.

Figure 8 shows how the Lane Management System handles the vehicle being outside of the speed threshold it operates within. The Lane Management System consistently checks if the vehicle speed is within acceptable limits. If the driver's speed is outside of these bounds the Lane Management System will send deactivate signals to all three subsystems, the Lane Keeping System, the Lane Departure Warning System, and the Lane Centering System. In response all three of these systems will deactivate.

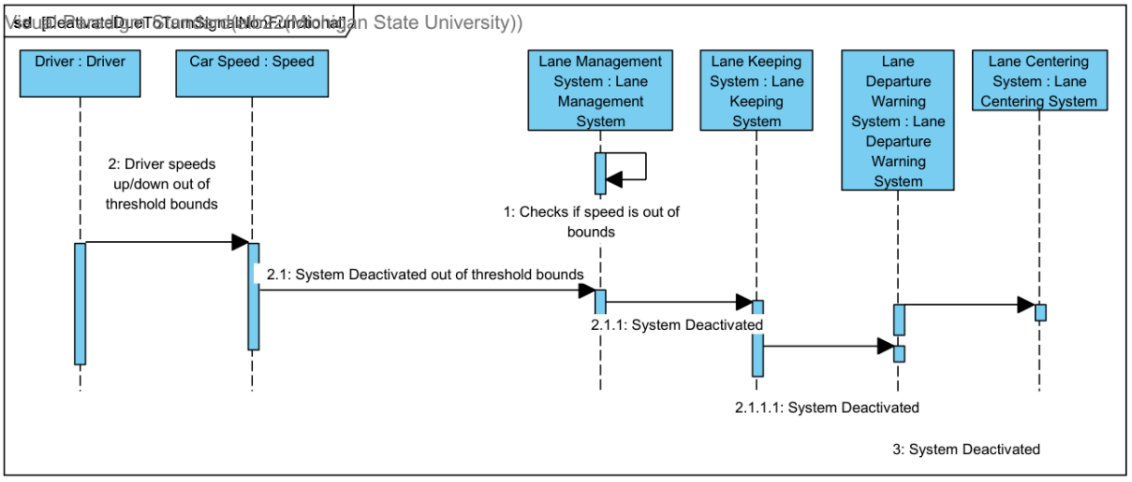


Figure 8: A sequence diagram showing the system handling when the driver is outside of the speed threshold that Lane Management System is active.

Figure 9 shows how the Lane Management System handles no lane markings being detected. The sequence starts by the camera sensors sending camera images to the image processing subsystem. The image processing subsystem sends the image data to the Lane Management System. The Lane Management System validates this camera data. If the camera data is invalid due to there being no lane markings the Lane Management System sends deactivate signals to all three subsystems, the Lane Departure Warning System, the Lane Keeping System, and the Lane Centering System. In response all three of these

systems deactivate.

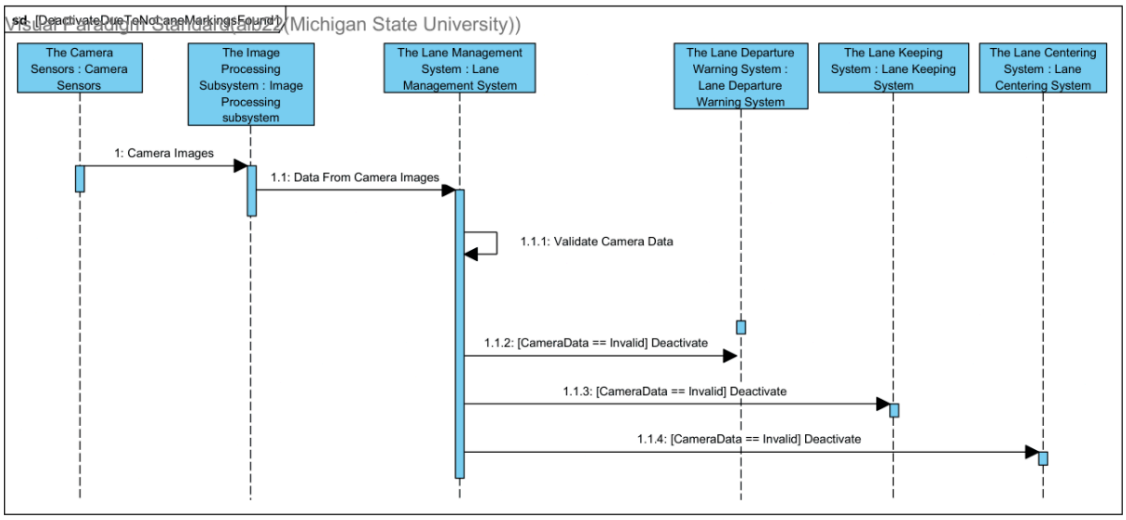


Figure 9: A sequence diagram showing the Lane Management System deactivating due to no lane markings found.

Figure 10 shows the normal case of the Lane Management System capturing and processing images. First the camera sensing subsystem runs camera diagnostics to ensure that the cameras are working properly. Next the camera sensing subsystem captures images of the road. The images are then sent to the image processing subsystem where they are processed to get the lane marking data. This data is sent to the path prediction subsystem. The path prediction subsystem processes the data then sends a message to the Lane Management containing the processed data. The Lane Management System activates its three subsystems to use as needed, the Lane Departure Warning System, the Lane Keeping System, and the Lane Centering System.

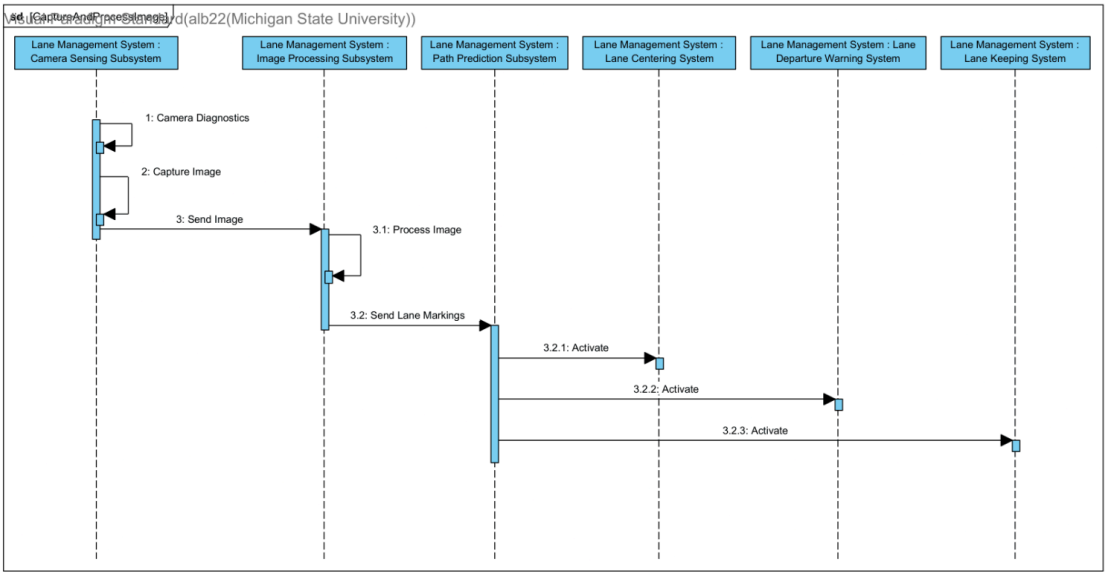


Figure 10: A sequence diagram showing the process of capturing camera data for the Lane Management

System to use.

Figure 11 shows the Lane Management System handling the exceptional case where the camera diagnostics fail. First the Lane Management System activates the camera sensing subsystem which runs the camera diagnostics. The camera sensing subsystem then sends a camera system failure signal back to the Lane Management System. After receiving this signal the Lane Management System deactivates by sending deactivate signals to its three subsystems, the Lane Departure Warning System, the Lane Keeping System, and the Lane Centering System.

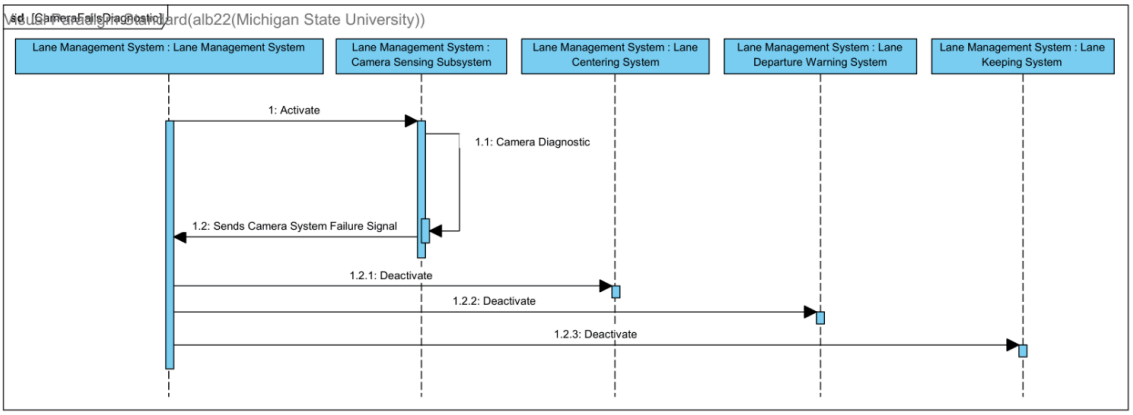


Figure 11: A sequence diagram showing the deactivation of the Lane Management System when the camera sensors fail.

Figure 12 shows how the Lane Management System can be overridden by the driver using the turn signal. First the driver turns on the turn signal. The turn signal then sends a message to the Lane Management System that it is active. After receiving this signal the Lane Management System sends deactivate signals to the Lane Keeping System and the Lane Departure Warning System. Both systems deactivate in response to these signals.

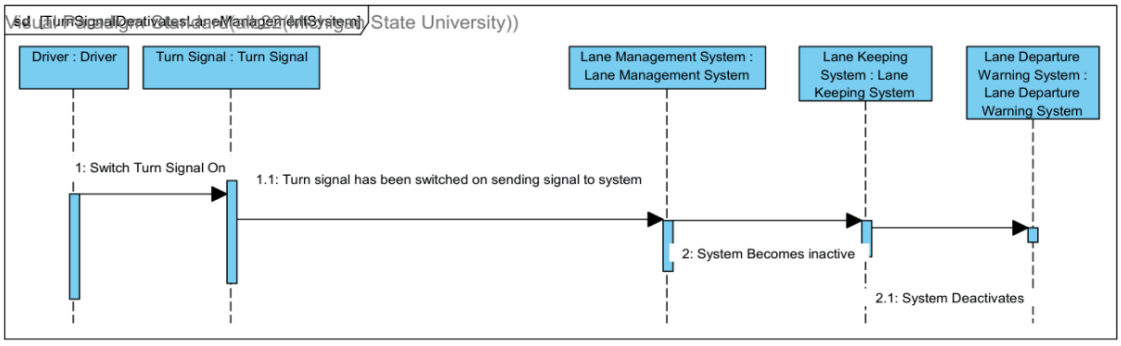


Figure 12: A sequence model showing turn signal deactivating the Lane Management System.

4.4 State Diagrams

Figure 13 shows a state diagram of the Image Processing System. The Image Processing System begins in the request image state where the Image Processing system requests images from the camera sensors. Once the images are received the Image Processing System moves into the process image state where the images will be processed into usable data. Once the images are processed the Image Processing System will move into the send image data state where the image data will be sent to many other classes for use. Finally the Image Processing System will wait then transition back to the request image state, restarting the loop of processing.

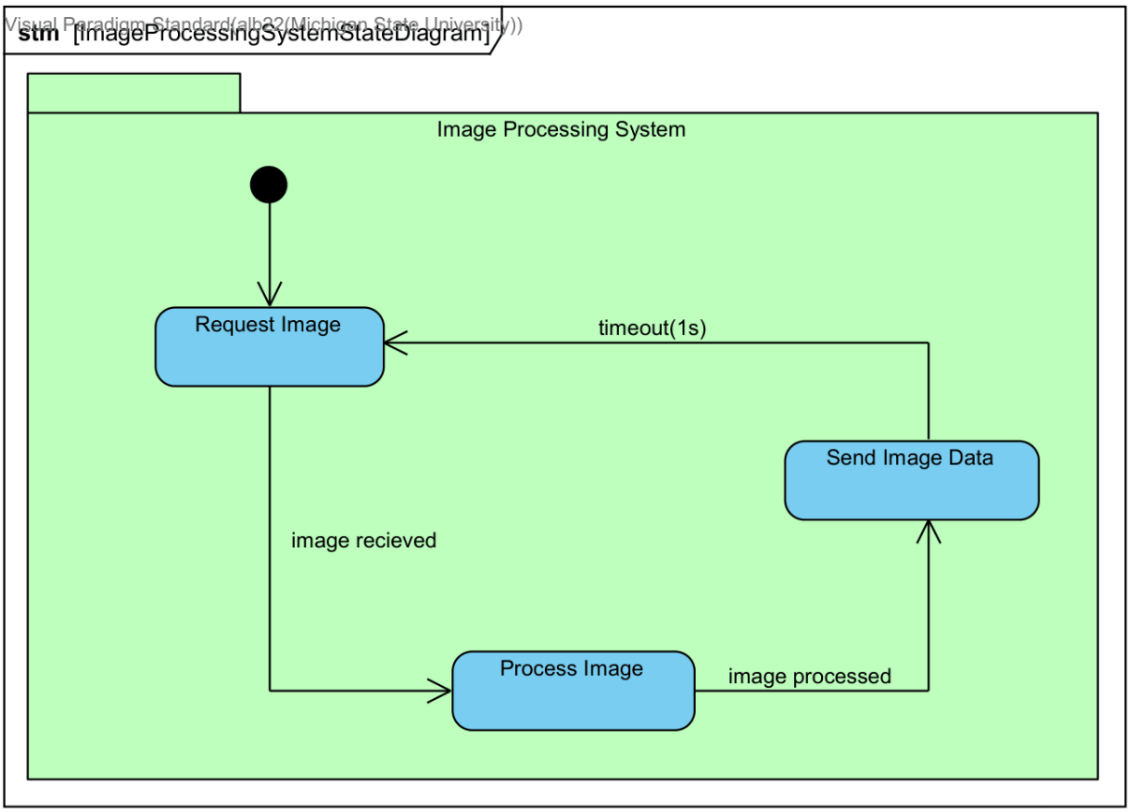


Figure 13: A state diagram of the Image Processing System.

Figure 14 shows a state diagram of the Lane Management System. The Lane Management System begins in the send active signals state. The Lane Management System will send activated signals to all three subsystems, the Lane Departure Warning System, the Lane Keeping System, and the Lane Centering System. Once all of those signals are sent the Lane Management System will transition to the enabled idle state. When in the enabled idle state if camera data is received, the Lane Management System will transition into the validate camera data state where the camera data will be checked to see if it is usable. If the camera data is valid the Lane Management System will transition back into the enabled idle state, if the camera data is invalid the Lane Management System will transition into the send deactivate signals state. Once every second the Lane Management System will transition into the check speed bounds state

where the system will check if the vehicle is moving at a speed within the bounds at which the Lane Management System requires to stay active. If the vehicle speed is within the bounds, the Lane Management System will transition back into the enabled idle state. If the speed is outside the bounds, the Lane Management System will transition to the send deactivate signals state. When in the send deactivate signals state the Lane Management System will send deactivate signals to all three subsystems, the Lane Departure Warning System, the Lane Keeping System, and the Lane Centering System. Once these signals are sent the Lane Management System will transition to the disabled idle state. If the system receives an activate signal the Lane Management System will transition into the send active signals state.

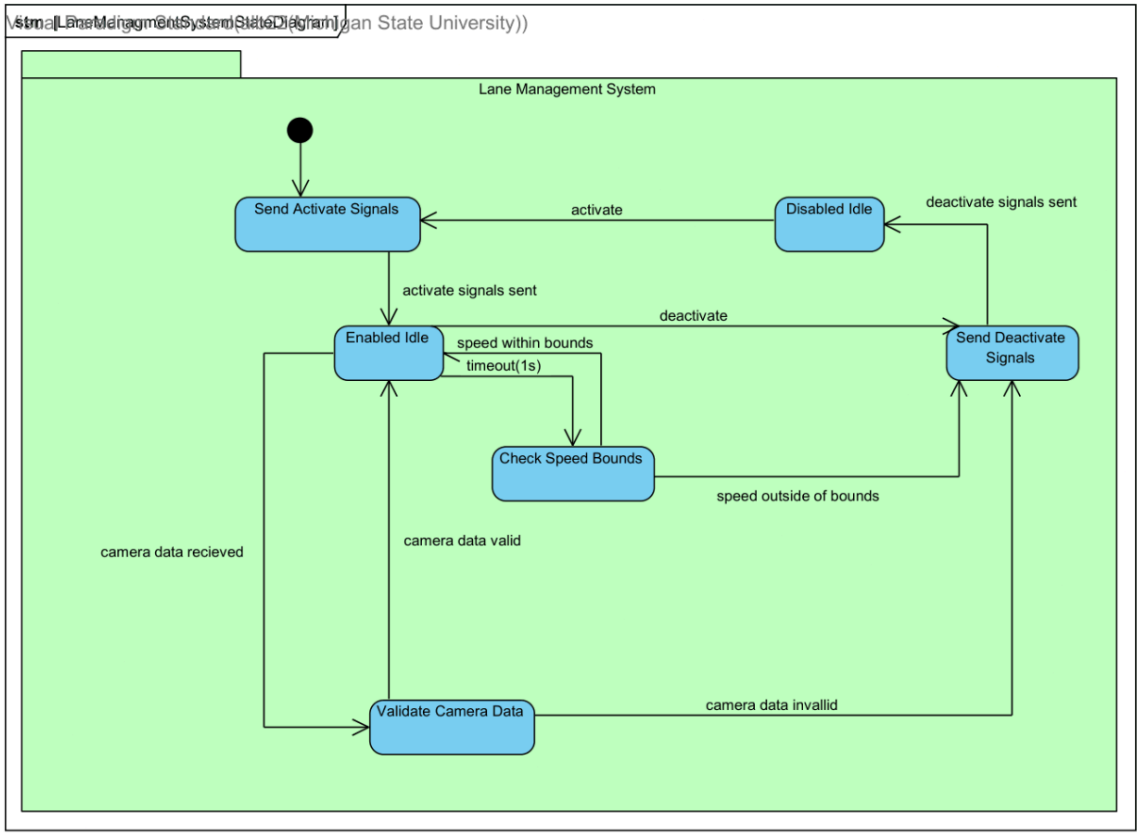


Figure 14: A state diagram of the Lane Management System.

Figure 15 shows a state diagram of the Path Prediction Subsystem. The Path Prediction Subsystem begins in the Idle state. Upon receiving a lane marker and the vehicle state it transitions into the Predicting state. Once the predicted path or a departure detection has been sent the Path Prediction Subsystem transitions back to the Idle state.

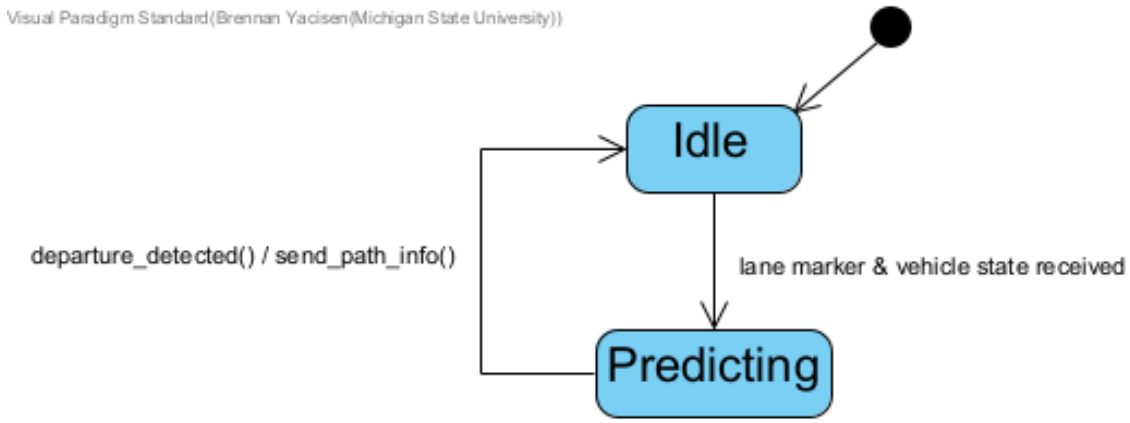


Figure 15: A state diagram of the path prediction subsystem.

Figure 16 shows a state diagram of the LKS. The LKS begins in the Idle state. Upon detection of a left lane departure the LKS transitions into the Keeping: Left state. Upon detection of a right lane departure the LKS transitions in the Keeping: Right state. From either the Keeping: Left state or the Keeping: Right the LKS transitions back to the Idle state if the vehicle is detected to be within the lane bounds.

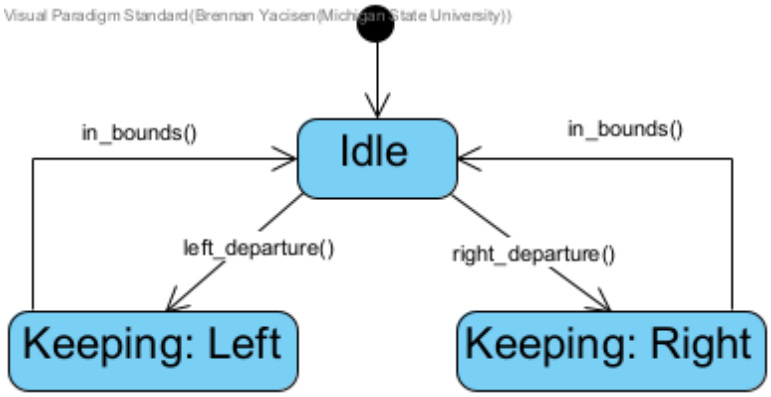


Figure 16: A state diagram of the LKS.

Figure 17 shows a state diagram of the turn signal. The turn signal begins in the Inactive state. The turn signal transitions to the Active: Left state when it is activated with direction = left, and it transitions to the Active: Right state when it is activated with direction = right. From either the Active: Left state or the Active: Right state the turn signal transitions to the Inactive state when it is deactivated.

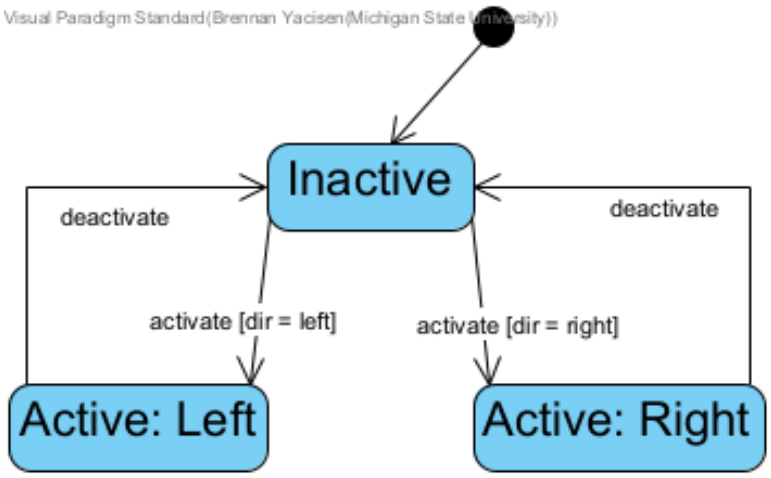


Figure 17: A state diagram of the turn signal.

Figure 18 shows a state diagram of the LDWS. It begins in its idle state and only activates when the Path Prediction System sends a warning. It starts by displaying the warning lights. If the Path Prediction System disables its warning then it returns to its idle state. If the car begins to depart the lane then it sends a message to activate the LKS. After the car has returned to the lane it goes back to its idle state and the cycle repeats.

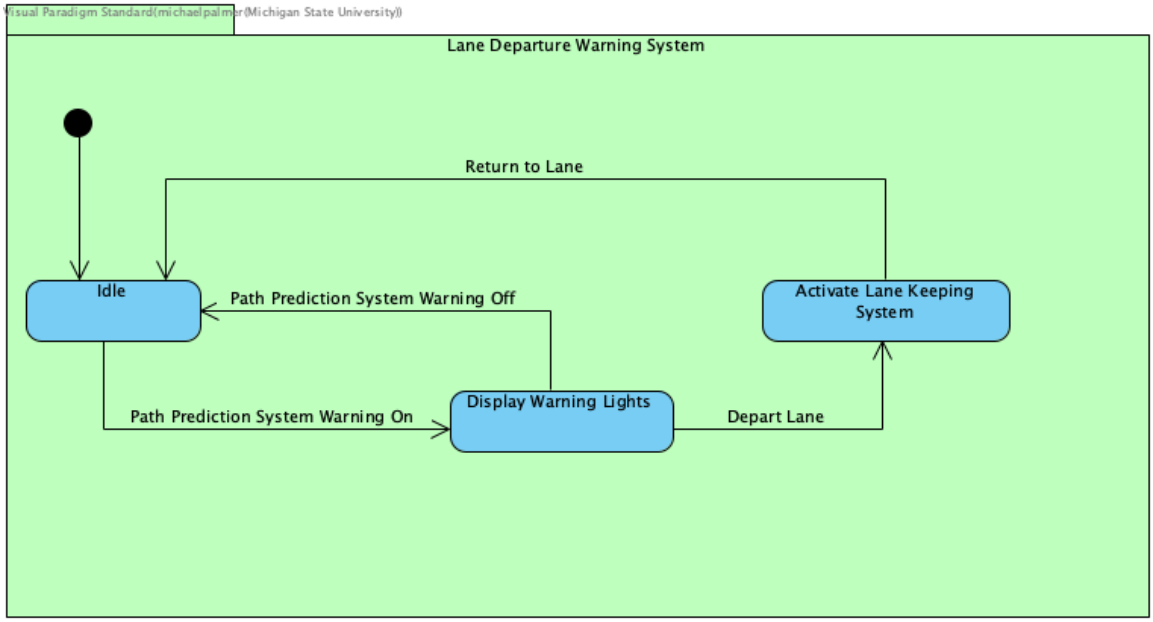


Figure 18: The LDWS state diagram.

Figure 19 shows a state diagram of the User Interface System. It starts by displaying the default interface and stays that way until the LMS sends a notification. The notification is then displayed on the interface for 5 seconds, then it returns to its default display.

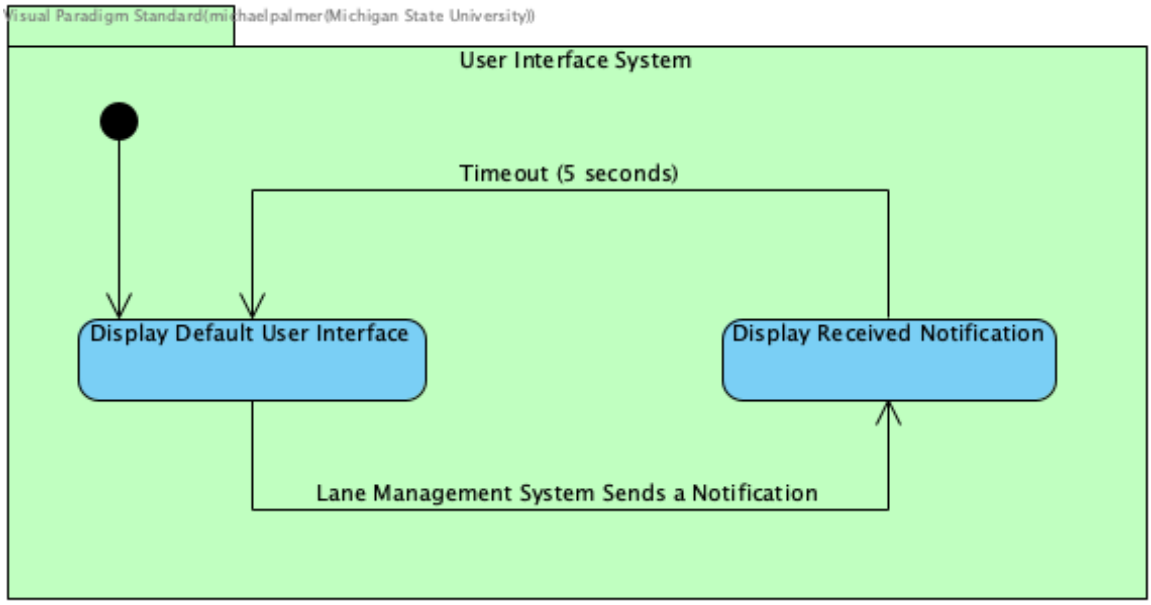


Figure 19: The User Interface System state diagram.

Figure 20 shows a diagram of the Lane Centering System. The system starts in the disabled state and then it transitions into the monitoring state on an "activate" signal, where it computes the vehicles distance offset from the center of the lane. If the required angle of steering has exceeded its maximum, the LCS transitions into the limit exceeded state and warns the driver to take control. The Lane Centering System transitions to the suspended state whenever the Lane Keeping System engages and goes back to monitoring once LKS disengages.

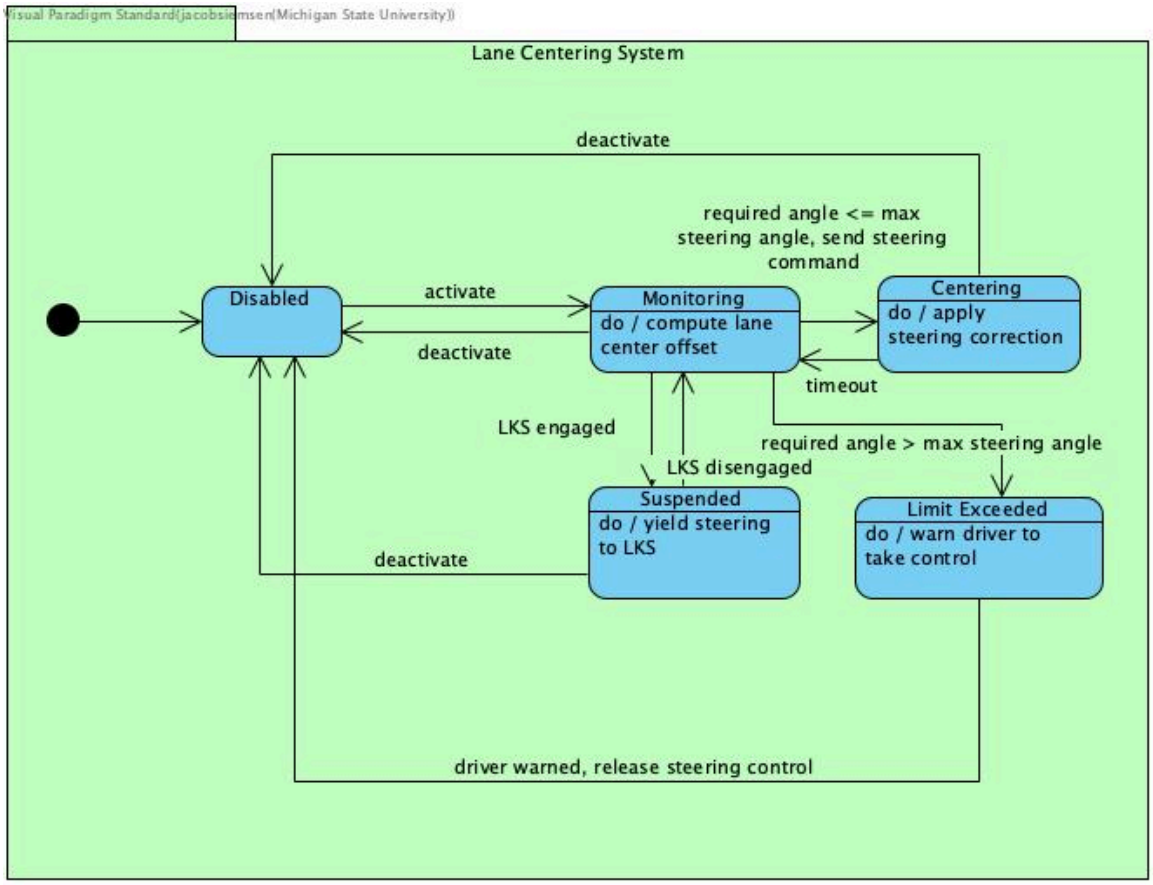


Figure 20: A state diagram of the Lane Centering System.

Figure 21 shows a diagram of the Steering Wheel, which is modeled as two independent state machines because haptic warnings and steering torque do not interact and the Steering Wheel has one state for each at the same time. It starts in the neutral state and transitions to the assisted steering state whenever torque is being applied from either the Lane Keeping System or Lane Centering System and returns once that torque force is removed. If the driver's torque they are applying to the Steering Wheel exceeds the threshold, an override is signaled and it transitions back into the Neutral state. The haptics subsystem starts in the Not Vibrating state and transitions to the Vibrating state once there is a lane departure warning and returns once the warning is cleared.

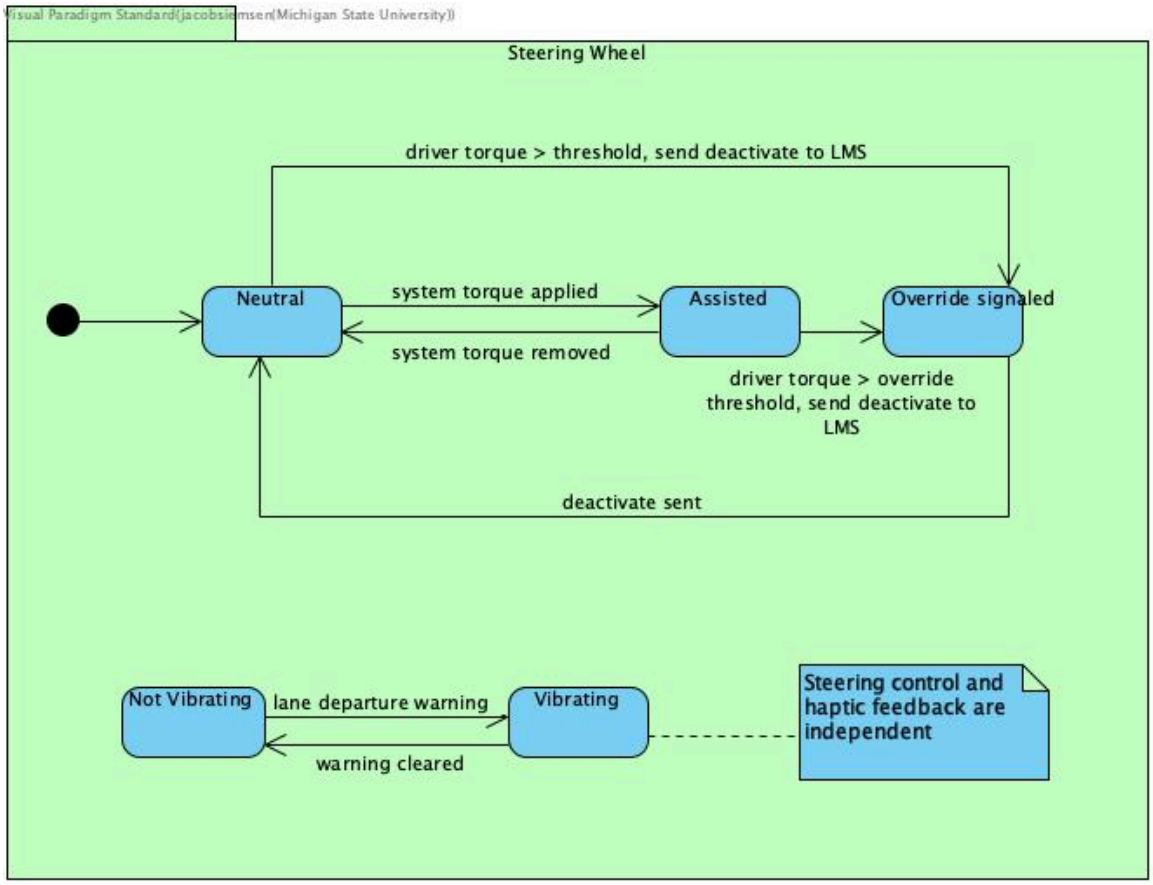


Figure 21: A state diagram of the Steering Wheel.

Figure 22 displays the state diagram of the driver. As the driver interacts with the Lane Management System, there are actions the driver may take that directly affect the state of the Lane Management System. The driver can decide to turn off this feature at any moment within their driving period. The driver may also choose to switch lanes or stay within one lane throughout the duration of their drive.

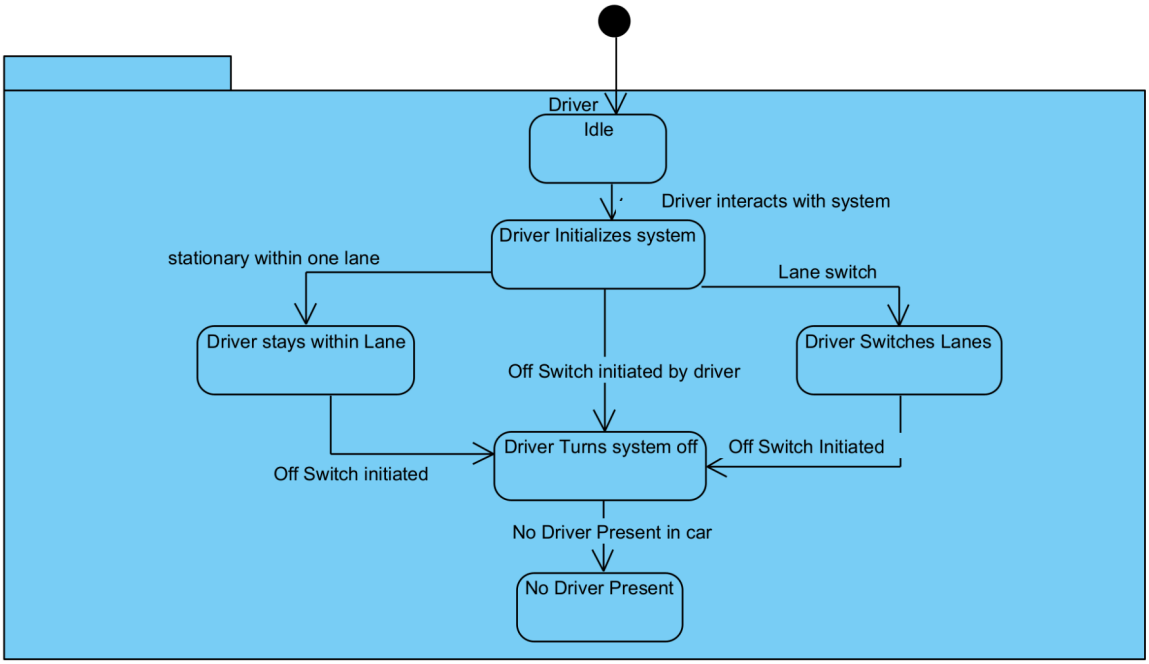


Figure 22: A State Diagram of the Driver.

Figure 23 showcases the state diagram for a Camera Sensor Subsystem. The sensor has different states based on the Lane Management System. Since the Camera sensor is a subsystem to the Lane Management System, it can be turned on and off as desired by the driver. The different states represented on the diagram display the camera sensor being active while driving, having a detection state if anything is detected, and a recovery state in the case of the sensor being unable to work as expected.

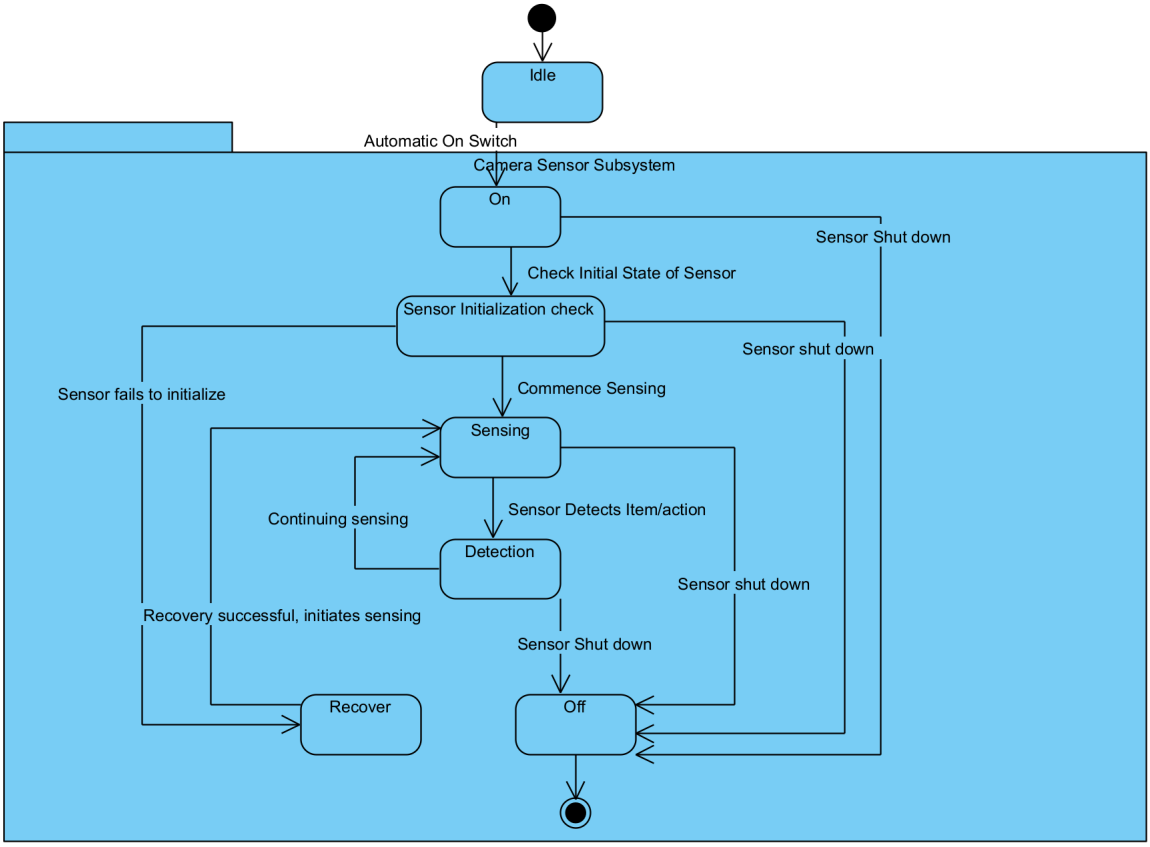


Figure 23: A State Diagram of the Camera Sensor System

5 Prototype

• Describe what your prototype will show in terms of system functionality.

5.1 How to Run Prototype

• Describe what is needed to run your prototype.

• What system configuration? (Should be accessible through web.) Are there plugins? Are there any OS or networking constraints? Give the URL for the prototype.

Prototype v1 does not have to interactive per se. But there should be sufficient number of interfaces (with descriptive captions/annotations) for the customer to understand the development team's interpretation of the requirements. All customer-provided scenarios should be included with the prototype.

Prototype V2 should also be accessible via a webpage. It should be executable and provide an interactive interface. All customer-provided scenarios should be supported and interactive with the prototype.

5.2 Sample Scenarios

• Give a sample scenario of using your system (from the set of scenarios provided by your customer). Use real data and problem scenarios. Include screen captures illustrating what your prototype produces. As always, be sure to describe all figures.

6 References

[1] “Lane keeping assist - my car does what,” My Car Does What -, https://www.mycardoeswhat.org/safety-features/lane-keeping-assist/

[2] “Lane Departure Warning - My Car does what,” My Car Does What -, https://www.mycardoeswhat.org/safety-features/lane-departure-warning/

[3] Bosch Mobility, https://www.bosch-mobility.com/en/solutions/assistance-systems/lane-centering-assist/

[4] “Lane-Centering Assist explained,” Capital One Auto Navigator, https://www.capitalone.com/cars/learn/finding-the-right-car/lanecentering-assist-explained/3838

[5] “CSE 435: Team Lane Management Systems 1,” Course Project Website, https://www.egr.msu.edu/cse435/2026/lms1/

7 Point of Contact

For further information regarding this document and project, please contact **Prof. Betty H.C. Cheng** at Michigan State University (chengb at msu.edu). All materials in this document have been sanitized for proprietary data. The students and the instructor gratefully acknowledge the participation of our industrial collaborators.